Technical Report: Positional Repeatability of Piston and Rail System

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1. Introduction and Objective

This report documents the results of repeatability tests conducted on a mechanical system comprising a **Piston** and a **Rail** (Stewart Platform) component. The primary **objective** of these tests was to quantify the positional accuracy and repeatability of the system by measuring the displacement of designated points after multiple movement cycles.

The study aims to establish the **Maximum Error** and **Maximum Deviation Range** to determine if the system's performance meets required engineering tolerances.

2. Methodology

2.1 Test Setup

Measurements were performed using a **Coordinate Measuring Machine (CMM)** to accurately record the coordinates **XYZ** of three calibrated reference points: **Point 1** (*PNT1)*, **Point 2** (*PNT2)*, and **Point 3** (*PNT3)* [Figure 2]

ie: **PNT1 (X_Point1, Y_Point1, Z_Point1), PNT2 (X_Point2, Y_Point2, Z_Point2),** and **PNT3 (X_Point3, Y_Point3, Z_Point3)**.

The system components were defined by their range of movement:

- **Piston:** Movement between **Top Dead Centre (TDC)** and **Bottom Dead Centre (BDC)**.
- **Rail/Platform:** Movement between **End 1** and **End 2**.
- **Stewart platform:** Stationary

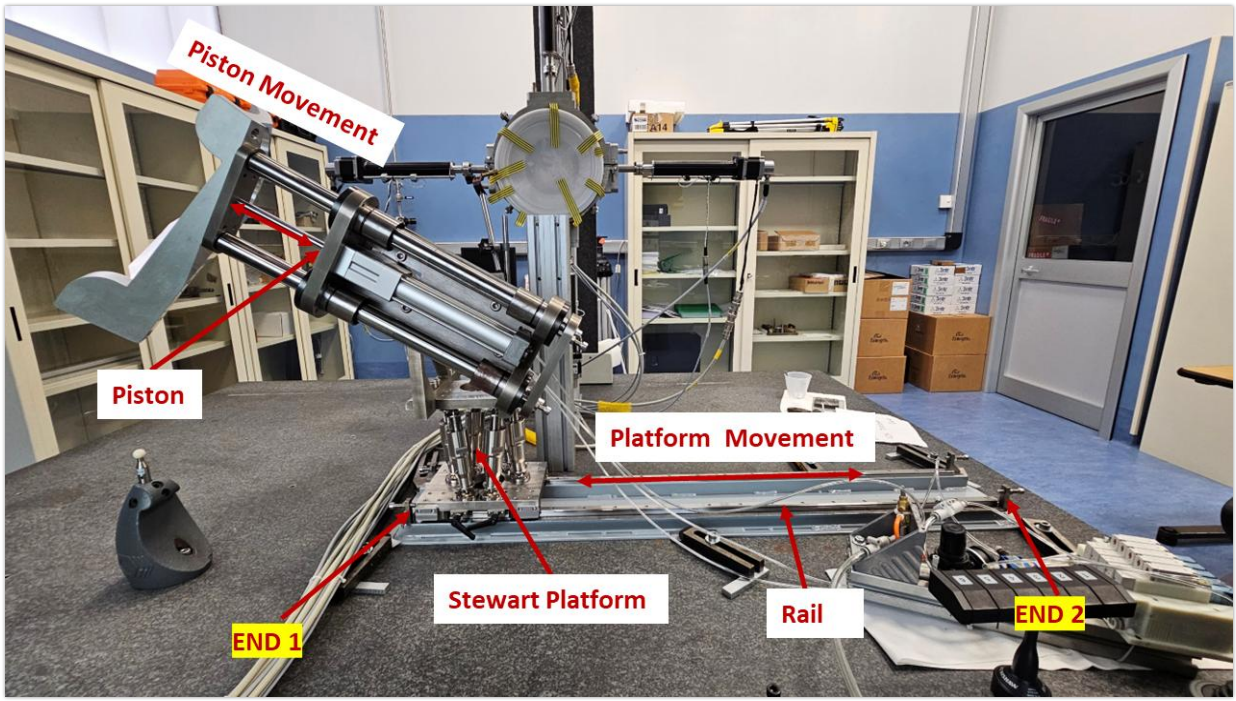


Figure 1: Right Octant Assembly

2.2 Test Procedures

Three primary test scenarios were executed:

A. Rail Repeatability Test (Piston Downside)

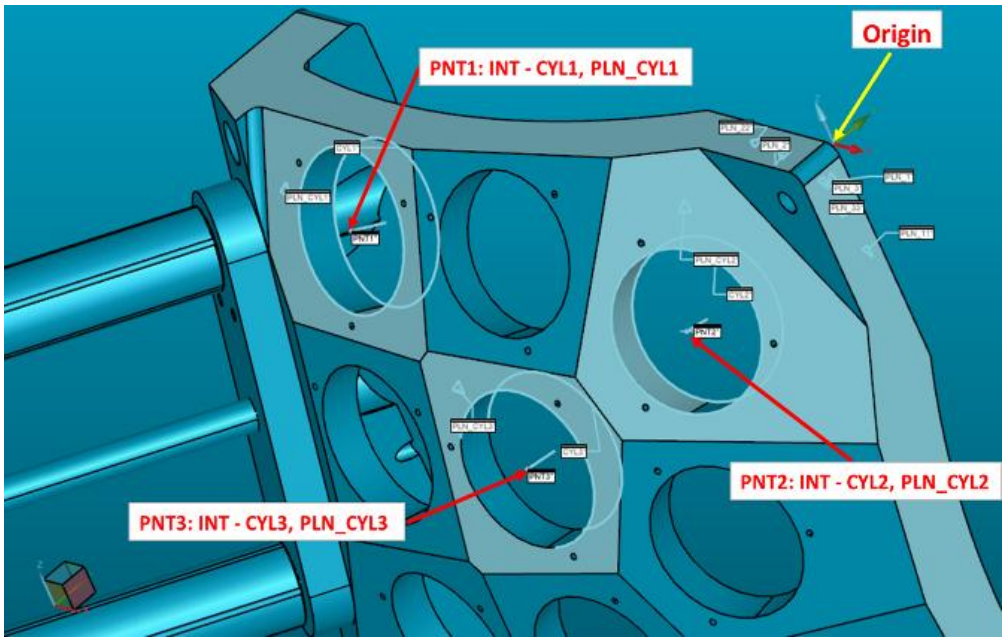


Figure 2: Shell indicating calibration Point1, Point2, Point3 and orgin for first calibration

PNT1: Intersection of Cylinder 1 and Plane of Cylinder 1 [Figure 2]

PNT2: Intersection of Cylinder 2 and Plane of Cylinder 2 [Figure 2]

PNT3: Intersection of Cylinder 3 and Plane of Cylinder 3 [Figure 2]

Initial Measurement: The rail was positioned at **End 1**, and the piston at **BDC** initial coordinates were recorded (with respect to origin*: Only for first test: Test1*).

1. **Movement Cycle:** The rail moved from **End 1** to **End 2** and then back to **End 1**.
2. **Repetition and Measurement:** Steps 1 and 2 were repeated **ten times** to gather ten independent measurements of **PNT1 (X_Point1, Y_Point1, Z_Point1), PNT2 (X_Point2, Y_Point2, Z_Point2),** and **PNT3 (X_Point3, Y_Point3, Z_Point3)** at the start position (End 1) after the movement cycle.

Test Number	TEST	X_Point1	Y_Point1	Z_Point1	X_Point2	Y_Point2	Z_Point2	X_Point3	Y_Point3	Z_Point3
Test1	Rail	-213.278	-67.254	-32.018	-65.743	-46.564	-59.545	-139.478	-84.678	-105.715
Test2	Rail	-213.241	-67.259	-32.019	-65.701	-46.594	-59.529	-139.438	-84.699	-105.708
Test3	Rail	-213.24	-67.248	-32.03	-65.697	-46.587	-59.538	-139.433	-84.691	-105.717
Test4	Rail	-213.239	-67.249	-32.028	-65.695	-46.589	-59.535	-139.43	-84.693	-105.715
Test5	Rail	-213.259	-67.25	-32.024	-65.72	-46.568	-59.544	-139.455	-84.679	-105.717
Test6	Rail	-213.28	-67.291	-31.982	-65.743	-46.607	-59.508	-139.478	-84.719	-105.68
Test7	Rail	-213.258	-67.255	-32.017	-65.721	-46.575	-59.537	-139.456	-84.685	-105.71
Test8	Rail	-213.24	-67.229	-32.042	-65.7	-46.56	-59.555	-139.433	-84.667	-105.732
Test9	Rail	-213.272	-67.231	-32.04	-65.736	-46.548	-59.563	-139.471	-84.662	-105.734
Test10	Rail	-213.263	-67.283	-31.994	-65.723	-46.616	-59.506	-139.458	-84.722	-105.683

B. Piston Repeatability Test (Rail Static)

1. **Initial Measurement:** The Piston was positioned at **TDC** (with the Rail held static), and the initial coordinates were recorded (with respect to origin*: Only for first test: Test1*).
2. **Movement Cycle:** The Piston moved from **TDC** to **BDC** and then back to **TDC**.
3. **Repetition and Measurement:** Steps 1 and 2 were repeated **ten times** to gather ten independent measurements of **PNT1 (X_Point1, Y_Point1, Z_Point1), PNT2 (X_Point2, Y_Point2, Z_Point2),** and **PNT3 (X_Point3, Y_Point3, Z_Point3)** at the start position (TDC) after the Piston movement cycle.

Test Number	TEST	X_Point1	Y_Point1	Z_Point1	X_Point2	Y_Point2	Z_Point2	X_Point3	Y_Point3	Z_Point3
Test1	Piston	-213.276	-67.295	-32.011	-65.743	-46.577	-59.546	-139.474	-84.706	-105.713
Test2	Piston	-213.28	-67.298	-32.008	-65.749	-46.577	-59.541	-139.48	-84.706	-105.71
Test3	Piston	-213.283	-67.3	-32.003	-65.751	-46.58	-59.537	-139.482	-84.707	-105.705
Test4	Piston	-213.284	-67.3	-32	-65.75	-46.578	-59.535	-139.481	-84.707	-105.703
Test5	Piston	-213.284	-67.302	-31.997	-65.752	-46.581	-59.531	-139.482	-84.709	-105.7
Test6	Piston	-213.287	-67.305	-31.992	-65.754	-46.581	-59.524	-139.485	-84.711	-105.695
Test7	Piston	-213.287	-67.305	-31.994	-65.754	-46.581	-59.525	-139.485	-84.71	-105.696
Test8	Piston	-213.281	-67.313	-31.988	-65.745	-46.594	-59.518	-139.476	-84.72	-105.687
Test9	Piston	-213.281	-67.313	-31.988	-65.747	-46.591	-59.52	-139.479	-84.718	-105.692
Test10	Piston	-213.282	-67.312	-31.987	-65.747	-46.591	-59.518	-139.479	-84.718	-105.69

C. Combined Piston and Rail Repeatability Test

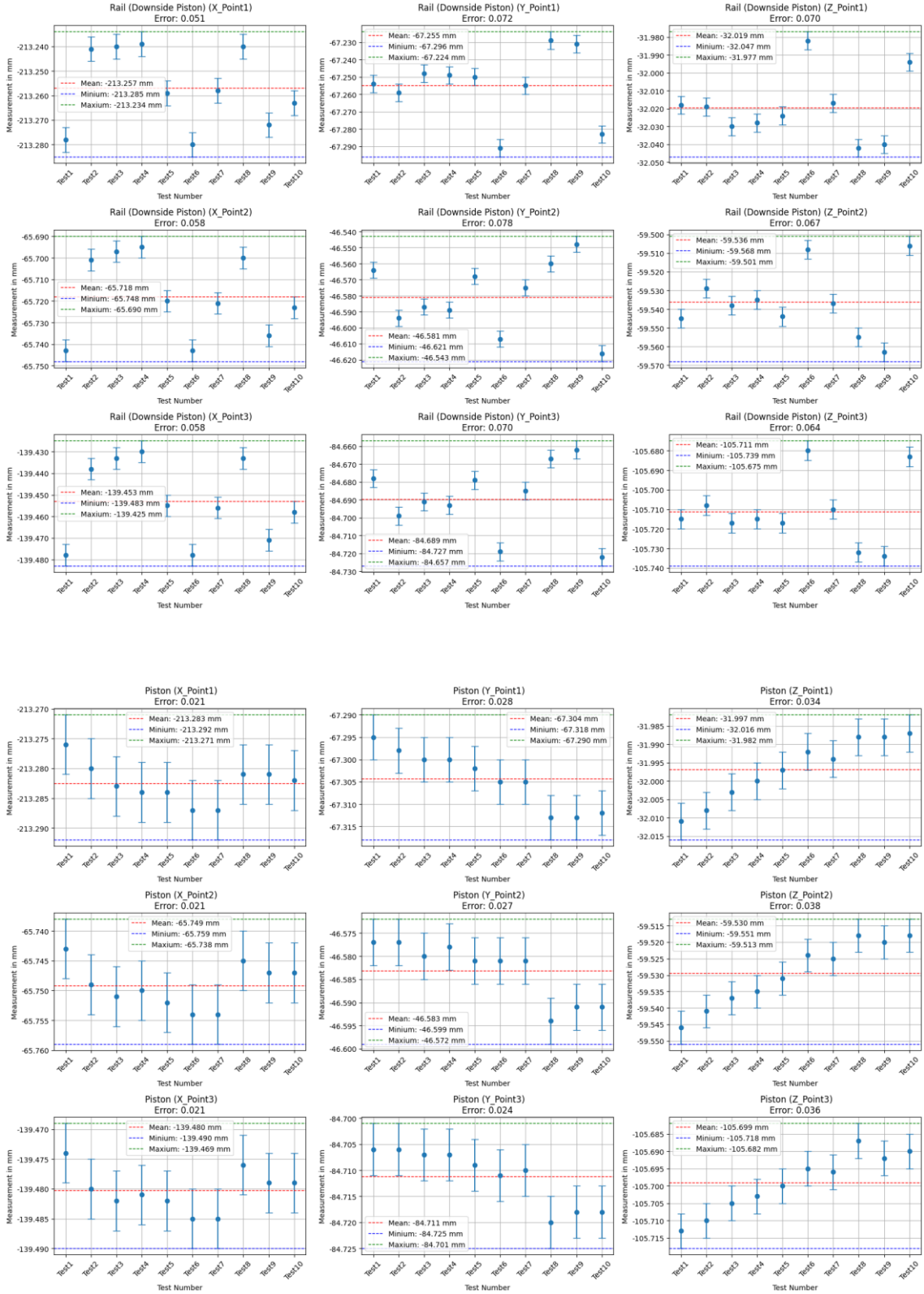
1. **Initial Measurement:** With the Piston at **TDC** and the Rail at **End 1**, the CMM recorded the initial coordinates of **PNT1 (X_Point1, Y_Point1, Z_Point1), PNT2 (X_Point2, Y_Point2, Z_Point2),** and **PNT3 (X_Point3, Y_Point3, Z_Point3)** (with respect to origin*: Only for first test: Test1*).
2. **Movement Cycle:**
   - Piston moved from TDC → BDC → TDC.
   - Rail moved from End 1 → End 2 → End 1.
3. **Repeat Measurement:** Coordinates were recorded again after the completion of the cycle.

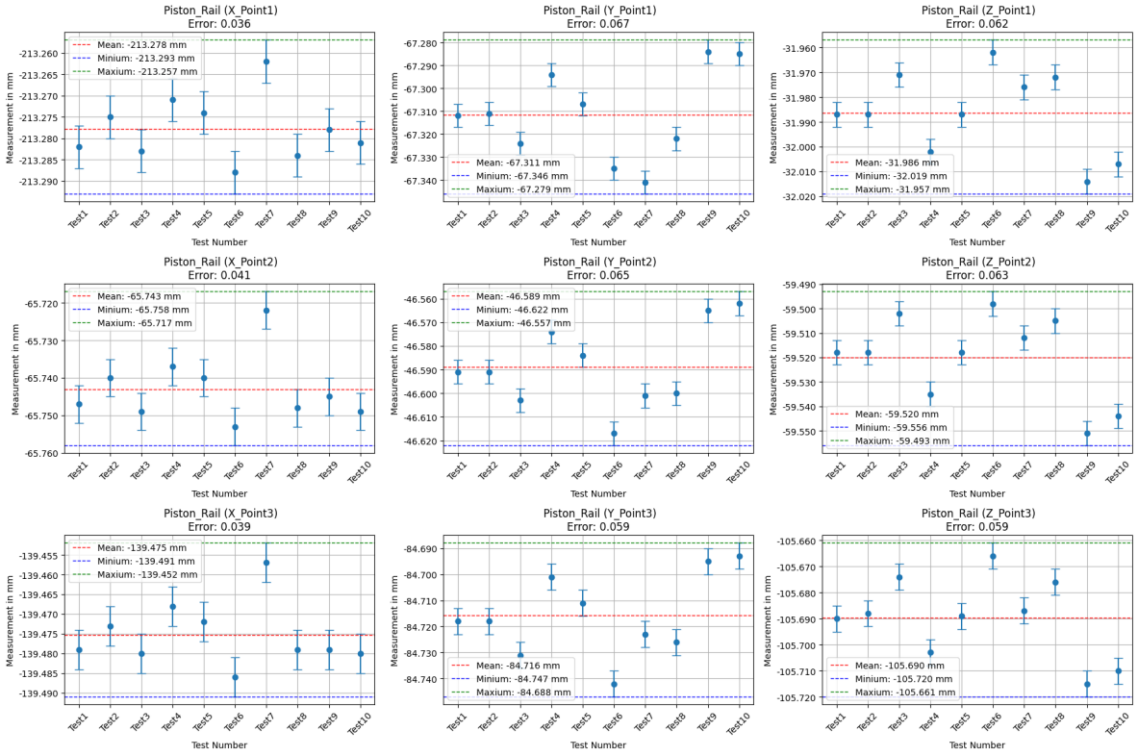
Test Number	TEST	X_Point1	Y_Point1	Z_Point1	X_Point2	Y_Point2	Z_Point2	X_Point3	Y_Point3	Z_Point3
Test1	Piston_Rail	-213.282	-67.312	-31.987	-65.747	-46.591	-59.518	-139.479	-84.718	-105.69
Test2	Piston_Rail	-213.275	-67.311	-31.987	-65.74	-46.591	-59.518	-139.473	-84.718	-105.688
Test3	Piston_Rail	-213.283	-67.324	-31.971	-65.749	-46.603	-59.502	-139.48	-84.731	-105.674
Test4	Piston_Rail	-213.271	-67.294	-32.002	-65.737	-46.574	-59.535	-139.468	-84.701	-105.703
Test5	Piston_Rail	-213.274	-67.307	-31.987	-65.74	-46.584	-59.518	-139.472	-84.711	-105.689
Test6	Piston_Rail	-213.288	-67.335	-31.962	-65.753	-46.617	-59.498	-139.486	-84.742	-105.666
Test7	Piston_Rail	-213.262	-67.341	-31.976	-65.722	-46.601	-59.512	-139.457	-84.723	-105.687
Test8	Piston_Rail	-213.284	-67.322	-31.972	-65.748	-46.6	-59.505	-139.479	-84.726	-105.676
Test9	Piston_Rail	-213.278	-67.284	-32.014	-65.745	-46.565	-59.551	-139.479	-84.695	-105.715
Test10	Piston_Rail	-213.281	-67.285	-32.007	-65.749	-46.562	-59.544	-139.48	-84.693	-105.71

3. Results and Conclusion

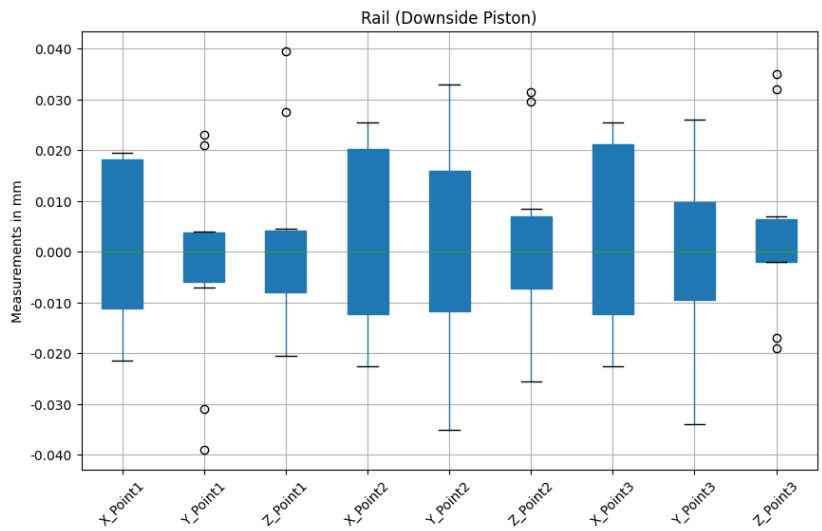
3.1 Key Findings

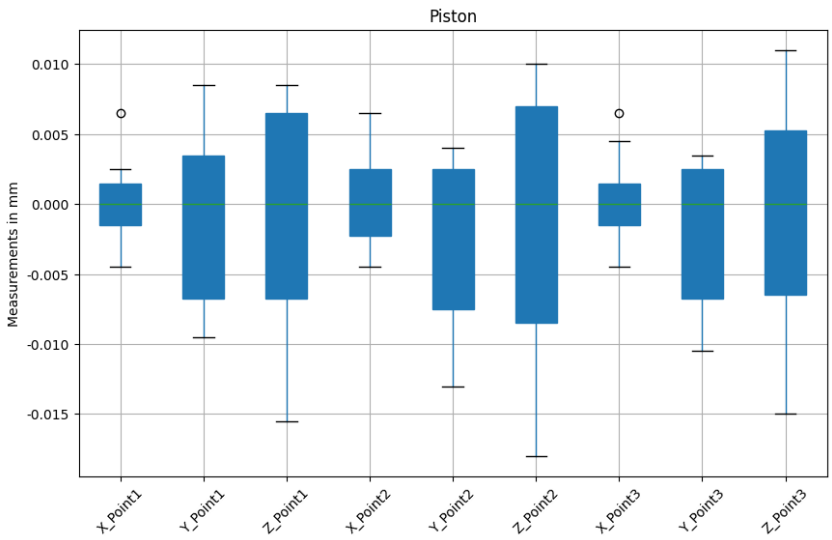
The collected data from the repeatability tests yielded the following critical figures:

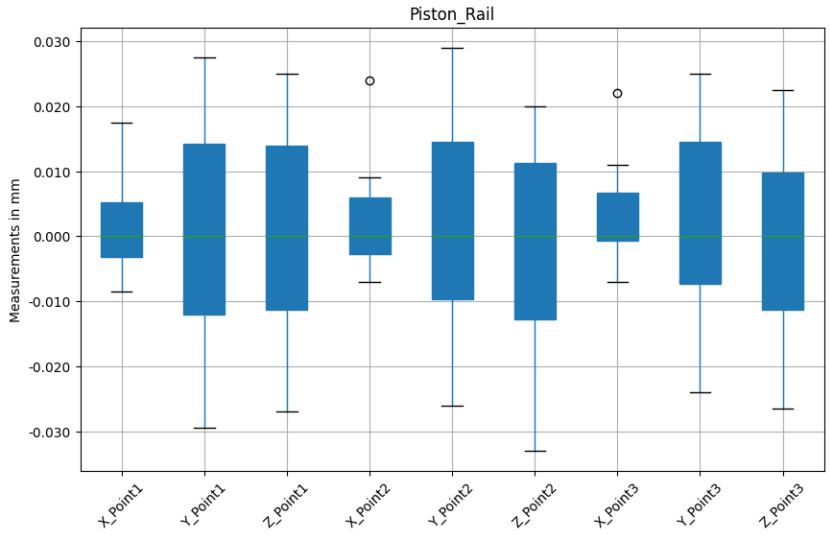




- **Maximum Error:** The largest recorded deviation from the initial reference position across all test points and cycles was determined to be 0.078 mm (± 5 micrometre machine tolerances included).
- **Maximum Deviation Range:** The overall range of positional fluctuation was observed to be less than ≤ ±0.04 mm from median (precise value redacted but indicating a very low tolerance range).







3.2 Conclusion

The repeatability tests confirm the positional stability of the mechanical system. The maximum recorded error of be 0.078 mm with deviation of ≤ ±0.04 mm from median which includes (machine tolerance of ± 5-micro meter) machine provides a key metric for assessing the system's precision. For critical applications, this value represents the certainty in the positioning of the system's end-effector after a full movement cycle.

Q&A

how do you obtain PNT1, 2 and 3?

They are the intersection of the inner cylinder of a hole and the corresponding external plane of the shell?

Ans: PNT1 (X_Point1, Y_Point1, Z_Point1), PNT2 (X_Point2, Y_Point2, Z_Point2), and PNT3 (X_Point3, Y_Point3, Z_Point3): PNT (X, Y, Z)

PNT1: Intersection of Cylinder 1 and Plane of Cylinder 1 [Figure 2]

PNT2: Intersection of Cylinder 2 and Plane of Cylinder 2 [Figure 2]

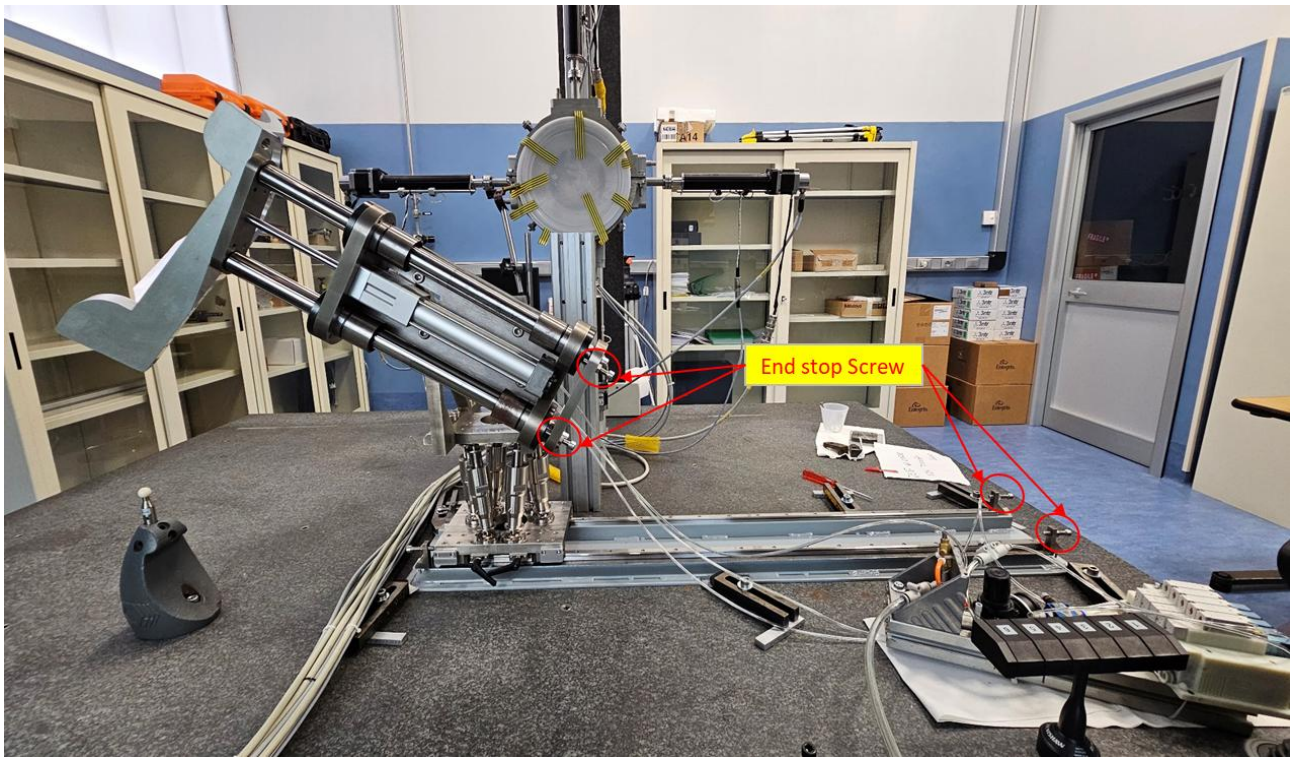
PNT3: Intersection of Cylinder 3 and Plane of Cylinder 3 [Figure 2]

Is it possible to know the ranges of movement of each component? Actuator (piston) and rail of the base. ?

What are the tolerances in the executive drawings, to which we have to compare the experimental measurements that you perform?

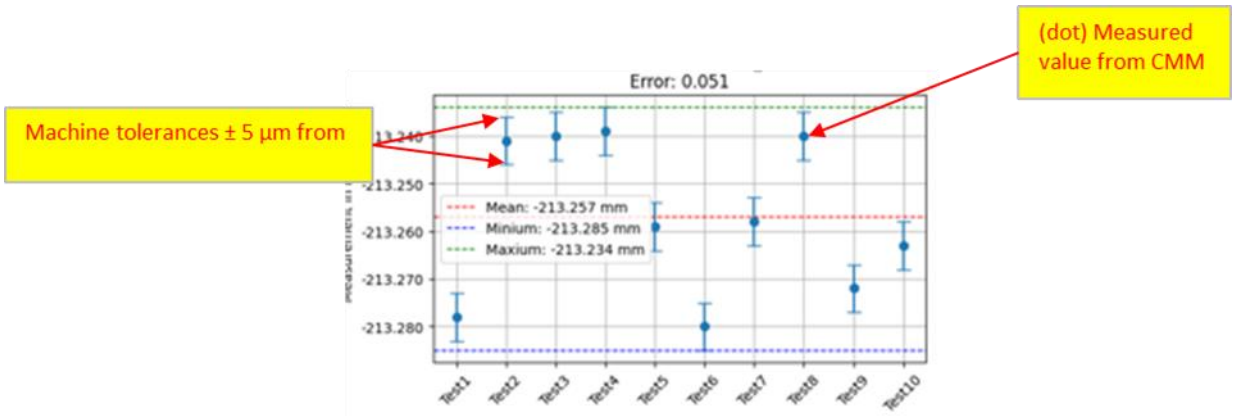
Ans: We can't have the range of moments for piston or rail, always they are moved to the extreme positions; we can adjust the movement by *End-stop screws* incorporated in rail and piston.

We can't have tolerances for assembly but only for individual components, that's why we do repeatability tests.



May you explain what is the first point X_Point1 for the Rail plot? What is the blue band and what is the error that you report?

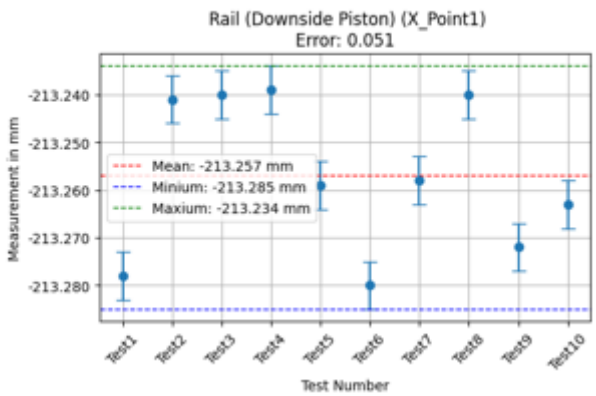
Ans: the blue dot is the CMM measured value, blue band refers machine tolerances ±5µm from measured value



The first point should be the mean of the point values of the first plot of page 4, with its error (difference max-min).

What is the initial reference position across all test points? It is not clear to me.

May you explain better?.



Ans: suppose for **Point 1, X-Coordinate** of Rail, we got 10 measurements of X, we plotted these 10 Points with machine tolerances ±5µm, mean is the average, maximum and minimum are extreme values. We don't have any true value or initial reference value. The main aim of repeatability test is the entire assembly will reach the same position after engaging and disengaging the gamma ray detectors.

I understood that all these measurements were obtained with a Stewart platform in the same condition, without moving any leg.

What about the work on this platform?

Ans: Stewart-platform was stationary with all legs at zero for ideal assembly conditions, but in real case scenario we use it for final adjustments (in 6DOF)

Do you have results of measurements and the program (also Matlab) to verify the precision in the movement of the upper plate, for example

or reference points?

Ans: No